

# Lab: Interdisciplinary Methods and Collaborative Research

**Learning Goal:** Practice designing interdisciplinary projects, communicating across domains, and implementing open science.

## Part 1: Interdisciplinary Project Design

**Exercise:** Design a research project that requires expertise from at least 3 disciplines:

**Example project:** "Active Inference models of anxiety in adolescents during social media use"

| Discipline                 | Contribution                  | Key Methods                                      |
|----------------------------|-------------------------------|--------------------------------------------------|
| Computational neuroscience | Generative model of anxiety   | POMDP specification, parameter fitting           |
| Developmental psychology   | Adolescent development theory | Longitudinal study design, age-appropriate tasks |
| Social media studies       | Platform behavior data        | Digital trace analysis, usage metrics            |
| Clinical psychology        | Assessment and intervention   | Clinical measures (GAD-7), therapeutic protocols |

**Project structure:**

1. Aim 1 (Computational): Develop Active Inference model of social media-induced anxiety
2. Aim 2 (Behavioral): Validate model against adolescent behavioral data
3. Aim 3 (Clinical): Use model to predict which adolescents are at risk and design interventions

Design your own 3-discipline project. Specify each discipline's contribution and how they integrate.

*Write your response here*

## Part 2: Communication Exercises

**Learning Goal:** Translate Active Inference for different audiences.

**Exercise:** Write a 150-word explanation of "precision-weighted prediction errors" for each audience:

**For a neuroscientist:** Focus on synaptic gain modulation, NMDA receptors, cortical layers, and oscillatory signatures. Use technical neuroscience terminology.

*Write your response here*

**For a clinical psychologist:** Focus on how confidence in beliefs vs. evidence determines interpretation. Use clinical examples (anxiety = too much weight on threat signals).

*Write your response here*

**For a high school student:** Use everyday analogies (confidence in what you expect to happen vs. what actually happens). Avoid all jargon.

*Write your response here*

Compare your three explanations. What was gained and lost in each translation?

*Write your response here*

## **Part 3: Open Science Implementation**

**Learning Goal:** Design a fully reproducible analysis.

**Exercise:** Design the reproducibility infrastructure for an Active Inference study:

| Component         | Implementation                               | Tool                               |
|-------------------|----------------------------------------------|------------------------------------|
| Pre-registration  | Register hypotheses, models, analysis plan   | OSF or AsPredicted                 |
| Version control   | All analysis code in Git repository          | GitHub/GitLab                      |
| Data sharing      | De-identified data in public repository      | OSF, Dryad, or OpenNeuro           |
| Model code        | Publish model specification as runnable code | Python package or Jupyter notebook |
| Environment       | Containerize computational environment       | Docker or conda environment.yml    |
| Documentation     | Literate programming with embedded analysis  | Jupyter notebooks + markdown       |
| Analysis pipeline | End-to-end automated pipeline                | Makefile or Snakemake              |

Write the README for this repository, including instructions for reproducing all results.

*Write your response here*

## Part 4: Workshop Design

**Learning Goal:** Design an educational workshop.

**Exercise:** Design a 2-day workshop introducing Active Inference to cognitive neuroscience graduate students:

## Day 1: Foundations

- Session 1 (2h): What is Active Inference? Core concepts without math (metaphors, intuitions)
- Session 2 (2h): The mathematics — free energy, generative models, policy selection
- Session 3 (2h): Hands-on: Building a simple Active Inference agent in pymdp

## Day 2: Applications

- Session 4 (2h): DCM — how to apply Active Inference to neuroimaging
- Session 5 (2h): Computational psychiatry — clinical applications
- Session 6 (2h): Design your own study — participants work in interdisciplinary teams

For each session, specify: learning objectives, materials needed, and assessment.

*Write your response here*

## Part 5: Reflection

In 300 words, reflect: Active Inference aspires to be a "theory of everything" for adaptive systems. But interdisciplinary ambition comes with risks — superficial application across domains, loss of disciplinary rigor, and over-promising. How do you balance the excitement of a unifying framework with the rigor needed for each specific application?

*Write your response here*

## Lab Summary

| Part | Skill Practiced          | Key Concept                    |
|------|--------------------------|--------------------------------|
| 1    | Project design           | Multi-disciplinary integration |
| 2    | Scientific communication | Audience adaptation            |
| 3    | Reproducibility          | Open science infrastructure    |
| 4    | Education                | Workshop curriculum design     |
| 5    | Critical reflection      | Ambition vs. rigor             |